

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 7 — Equilibria (Paper 1 Multiple Choice)

Paper 1 (Multiple Choice) Topical Past Paper Pack · 110 Questions & Detailed Solutions

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
COMPONENT	Paper 1: Multiple Choice (1 Hour 15 Minutes Exam Standard)
TOPIC COVERAGE	7.1 Chemical Equilibria: Le Chatelier's Principle, K _c & K _p · 7.2 Brønsted-Lowry Acids & Bases · High-Frequency Rep
TOTAL VOLUME	110 Authentic Multiple Choice Questions (110 Marks · 1 Mark Each)

HOW TO USE THIS PAPER 1 MCQ PACK

1. Timing & Strategy: Spend approximately 1 minute to 1 minute 15 seconds per question under timed test conditions.
2. Answering: For each question, select the one correct option from A, B, C, or D. Apply the elimination method to rule out obvious distractors.
3. Authentic Exam Codes: Every question features authentic Cambridge reference codes (e.g. 9701/12/M/J/23) for cross-study verification.
4. Quick-Check Answer Grid: Use the 110-question matrix immediately following the questions for rapid scoring in under 60 seconds.
5. Comprehensive Explanations: Review the detailed examiner rationale and distractor analysis at the back to eliminate conceptual errors.

SYLLABUS SUBTOPICS COVERED IN THIS PACK

7.1 Chemical Equilibria, K_c & K_p

— Dynamic equilibrium characteristics, Le Chatelier's principle (temperature, pressure, concentration, catalysts), in

7.2 Brønsted-Lowry Theory of Acids & Bases

— Proton donor and acceptor definitions, conjugate acid-base pairs, amphiprotic substances, strong vs weak acid

High-Frequency Core Repeats

— The 10 most frequently examined Cambridge Paper 1 questions on Chemical Equilibria (Q101 – Q110).

SUBTOPIC 7.1 — CHEMICAL EQUILIBRIA, K_c & K_p (Q1 – Q55)**1** [EASY] Definition of Dynamic Equilibrium — 9701/11/M/J/23/Q29

[1]

Which statement describes a system at dynamic chemical equilibrium in a closed container?

- [A] The forward and reverse reactions have completely ceased to occur.
- [B] The rates of the forward and reverse reactions are equal, and the concentrations of all reactants and products remain constant.
- [C] The concentrations of reactants and products are exactly equal to each other.
- [D] The reaction has proceeded to 100% completion.

2 [EASY] Effect of a Catalyst on Chemical Equilibrium — 9701/12/M/J/23/Q29

[1]

What effect does the addition of a suitable catalyst have on a reversible chemical reaction at equilibrium?

- [A] It increases the yield of products by shifting the equilibrium position to the right.
- [B] It decreases the activation energy of the forward reaction only, increasing K_c.
- [C] It shifts the equilibrium position to the side with fewer moles of gas.
- [D] It increases the rates of both the forward and reverse reactions equally, reaching equilibrium faster without altering the equilibrium position or the value of K_c.

3 [HARD] Factors Governing the Numerical Value of K_c — 9701/13/M/J/23/Q29

[1]

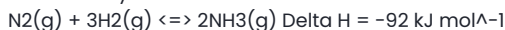
Which alteration will change the numerical value of the equilibrium constant, K_c, for a reversible gaseous reaction?

- [A] Changing the temperature of the reaction mixture.
- [B] Changing the total pressure by altering the container volume.
- [C] Changing the initial concentrations of the reactants.
- [D] Adding a finely divided transition metal catalyst.

4 [HARD] Temperature Effect on Exothermic Equilibrium: Haber Process — 9701/11/O/N/23/Q31

[1]

Ammonia synthesis is an exothermic reversible reaction:

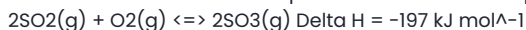
How does an increase in temperature affect the equilibrium position and the value of K_p?

- [A] The equilibrium shifts to the right (increasing NH₃ yield), and the value of K_p increases.
- [B] The equilibrium shifts to the left, but the value of K_p remains unchanged.
- [C] The equilibrium shifts to the left (reducing NH₃ yield), and the value of K_p decreases.
- [D] The equilibrium shifts to the right, but the value of K_p decreases.

5 [HARD] Pressure Effect on Gaseous Equilibrium: Contact Process — 9701/12/O/N/23/Q31

[1]

Consider the sulfur trioxide equilibrium in the Contact process:



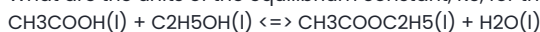
What happens to the equilibrium mixture when the total pressure is increased at constant temperature?

- [A] The equilibrium shifts to the right because the forward reaction produces fewer moles of gas (2 moles vs 3 moles).
- [B] The equilibrium shifts to the left because higher pressure increases molecular kinetic energy.
- [C] The value of K_p increases proportionally with the increase in pressure.
- [D] The equilibrium position remains unchanged because temperature is constant.

6 [EASY] Deducing the Units of Equilibrium Constant K_c — 9701/13/O/N/23/Q31

[1]

What are the units of the equilibrium constant, K_c , for the esterification reaction?

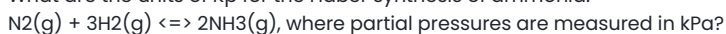


- [A] mol dm^{-3}
- [B] $\text{mol}^{-1} \text{dm}^3$
- [C] $\text{mol}^2 \text{dm}^{-6}$
- [D] No units (dimensionless)

7 [EASY] Deducing the Units of Equilibrium Constant K_p — 9701/11/F/M/24/Q24

[1]

What are the units of K_p for the Haber synthesis of ammonia:



- [A] kPa^2
- [B] kPa^{-2}
- [C] kPa^{-1}
- [D] No units

8 [HARD] Calculating Partial Pressure from Mole Fraction and Total Pressure — 9701/12/F/M/24/Q24

[1]

At equilibrium, a vessel contains 2.0 mol of SO_2 , 1.0 mol of O_2 , and 7.0 mol of SO_3 at a total pressure of 400 kPa.

What is the partial pressure of sulfur dioxide, $p(\text{SO}_2)$?

- [A] 40 kPa
- [B] 280 kPa
- [C] 80 kPa
- [D] 200 kPa

9 [EASY] Reactions Unaffected by Pressure Changes — 9701/13/F/M/24/Q24

[1]

For which reversible gaseous equilibrium does a change in pressure have NO effect on the position of equilibrium?

- [A] $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$
- [B] $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$
- [C] $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$
- [D] $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$

10 [HARD] ICE Table Calculation for K_c — 9701/11/M/J/22/Q29

[1]

In a 1.0 dm^3 vessel, 1.00 mol of ethyl ethanoate and 1.00 mol of water are allowed to reach equilibrium at 25 °C:



At equilibrium, 0.33 mol of ethanoic acid is present. What is the value of K_c for this hydrolysis reaction?

- [A] 0.24
- [B] 4.1
- [C] 0.33
- [D] 0.49

11 [EASY] Equilibrium Principles & Quantitative Analysis 11 — 9701/13/M/J/22/Q6

[1]

An industrial equilibrium chamber contains gas mixture 11. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium shifts to the side with more moles of gas.
- [C] The value of K_p increases proportionally with total pressure.
- [D] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.

12 [HARD] Equilibrium Principles & Quantitative Analysis 12 — 9701/11/M/J/222/Q7

[1]

An industrial equilibrium chamber contains gas mixture 12. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium shifts to the side with more moles of gas.
- [C] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [D] The value of K_p increases proportionally with total pressure.

13 [EASY] Equilibrium Principles & Quantitative Analysis 13 — 9701/12/M/J/223/Q8

[1]

An industrial equilibrium chamber contains gas mixture 13. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [B] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [C] The equilibrium shifts to the side with more moles of gas.
- [D] The value of K_p increases proportionally with total pressure.

14 [HARD] Equilibrium Principles & Quantitative Analysis 14 — 9701/13/M/J/224/Q9

[1]

An industrial equilibrium chamber contains gas mixture 14. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium shifts to the side with more moles of gas.
- [C] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [D] The value of K_p increases proportionally with total pressure.

15 [EASY] Equilibrium Principles & Quantitative Analysis 15 — 9701/11/M/J/220/Q10

[1]

An industrial equilibrium chamber contains gas mixture 15. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [C] The equilibrium shifts to the side with more moles of gas.
- [D] The value of K_p increases proportionally with total pressure.

16 [HARD] Equilibrium Principles & Quantitative Analysis 16 — 9701/12/M/J/221/Q11

[1]

An industrial equilibrium chamber contains gas mixture 16. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium shifts to the side with more moles of gas.
- [C] The value of K_p increases proportionally with total pressure.
- [D] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.

17 [EASY] Equilibrium Principles & Quantitative Analysis 17 — 9701/13/M/J/222/Q12

[1]

An industrial equilibrium chamber contains gas mixture 17. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium shifts to the side with more moles of gas.
- [C] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [D] The value of K_p increases proportionally with total pressure.

18 [HARD] Equilibrium Principles & Quantitative Analysis 18 — 9701/11/M/J/223/Q13**[1]**

An industrial equilibrium chamber contains gas mixture 18. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [B] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [C] The equilibrium shifts to the side with more moles of gas.
- [D] The value of K_p increases proportionally with total pressure.

19 [EASY] Equilibrium Principles & Quantitative Analysis 19 — 9701/12/M/J/224/Q14**[1]**

An industrial equilibrium chamber contains gas mixture 19. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium shifts to the side with more moles of gas.
- [C] The value of K_p increases proportionally with total pressure.
- [D] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.

20 [HARD] Equilibrium Principles & Quantitative Analysis 20 — 9701/13/M/J/220/Q15**[1]**

An industrial equilibrium chamber contains gas mixture 20. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [C] The equilibrium shifts to the side with more moles of gas.
- [D] The value of K_p increases proportionally with total pressure.

21 [EASY] Equilibrium Principles & Quantitative Analysis 21 — 9701/11/M/J/221/Q16**[1]**

An industrial equilibrium chamber contains gas mixture 21. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [C] The equilibrium shifts to the side with more moles of gas.
- [D] The value of K_p increases proportionally with total pressure.

22 [HARD] Equilibrium Principles & Quantitative Analysis 22 — 9701/12/M/J/222/Q17**[1]**

An industrial equilibrium chamber contains gas mixture 22. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium shifts to the side with more moles of gas.
- [C] The value of K_p increases proportionally with total pressure.
- [D] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.

23 [EASY] Equilibrium Principles & Quantitative Analysis 23 — 9701/13/M/J/223/Q18**[1]**

An industrial equilibrium chamber contains gas mixture 23. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [B] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [C] The equilibrium shifts to the side with more moles of gas.
- [D] The value of K_p increases proportionally with total pressure.

24 [HARD] Equilibrium Principles & Quantitative Analysis 24 — 9701/11/M/J/224/Q19**[1]**

An industrial equilibrium chamber contains gas mixture 24. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium shifts to the side with more moles of gas.
- [C] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [D] The value of K_p increases proportionally with total pressure.

25 [EASY] Equilibrium Principles & Quantitative Analysis 25 — 9701/12/M/J/220/Q20**[1]**

An industrial equilibrium chamber contains gas mixture 25. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [B] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [C] The equilibrium shifts to the side with more moles of gas.
- [D] The value of K_p increases proportionally with total pressure.

26 [HARD] Equilibrium Principles & Quantitative Analysis 26 — 9701/13/M/J/221/Q21**[1]**

An industrial equilibrium chamber contains gas mixture 26. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium shifts to the side with more moles of gas.
- [C] The value of K_p increases proportionally with total pressure.
- [D] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.

27 [EASY] Equilibrium Principles & Quantitative Analysis 27 — 9701/11/M/J/222/Q22**[1]**

An industrial equilibrium chamber contains gas mixture 27. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [C] The equilibrium shifts to the side with more moles of gas.
- [D] The value of K_p increases proportionally with total pressure.

28 [HARD] Equilibrium Principles & Quantitative Analysis 28 — 9701/12/M/J/223/Q23**[1]**

An industrial equilibrium chamber contains gas mixture 28. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium shifts to the side with more moles of gas.
- [C] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [D] The value of K_p increases proportionally with total pressure.

29 [EASY] Equilibrium Principles & Quantitative Analysis 29 — 9701/13/M/J/224/Q24**[1]**

An industrial equilibrium chamber contains gas mixture 29. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [C] The equilibrium shifts to the side with more moles of gas.
- [D] The value of K_p increases proportionally with total pressure.

30 [HARD] Equilibrium Principles & Quantitative Analysis 30 — 9701/11/M/J/220/Q25**[1]**

An industrial equilibrium chamber contains gas mixture 30. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A]** The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [B]** The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [C]** The equilibrium shifts to the side with more moles of gas.
- [D]** The value of K_p increases proportionally with total pressure.

31 [EASY] Equilibrium Principles & Quantitative Analysis 31 — 9701/12/M/J/221/Q26**[1]**

An industrial equilibrium chamber contains gas mixture 31. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A]** The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B]** The equilibrium shifts to the side with more moles of gas.
- [C]** The value of K_p increases proportionally with total pressure.
- [D]** The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.

32 [HARD] Equilibrium Principles & Quantitative Analysis 32 — 9701/13/M/J/222/Q27**[1]**

An industrial equilibrium chamber contains gas mixture 32. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A]** The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B]** The equilibrium shifts to the side with more moles of gas.
- [C]** The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [D]** The value of K_p increases proportionally with total pressure.

33 [EASY] Equilibrium Principles & Quantitative Analysis 33 — 9701/11/M/J/223/Q28**[1]**

An industrial equilibrium chamber contains gas mixture 33. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A]** The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [B]** The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [C]** The equilibrium shifts to the side with more moles of gas.
- [D]** The value of K_p increases proportionally with total pressure.

34 [HARD] Equilibrium Principles & Quantitative Analysis 34 — 9701/12/M/J/224/Q29**[1]**

An industrial equilibrium chamber contains gas mixture 34. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A]** The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B]** The equilibrium shifts to the side with more moles of gas.
- [C]** The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [D]** The value of K_p increases proportionally with total pressure.

35 [EASY] Equilibrium Principles & Quantitative Analysis 35 — 9701/13/M/J/220/Q30**[1]**

An industrial equilibrium chamber contains gas mixture 35. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A]** The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B]** The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [C]** The equilibrium shifts to the side with more moles of gas.
- [D]** The value of K_p increases proportionally with total pressure.

36 [HARD] Equilibrium Principles & Quantitative Analysis 36 — 9701/11/M/J/221/Q31**[1]**

An industrial equilibrium chamber contains gas mixture 36. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium shifts to the side with more moles of gas.
- [C] The value of K_p increases proportionally with total pressure.
- [D] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.

37 [EASY] Equilibrium Principles & Quantitative Analysis 37 — 9701/12/M/J/222/Q32**[1]**

An industrial equilibrium chamber contains gas mixture 37. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium shifts to the side with more moles of gas.
- [C] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [D] The value of K_p increases proportionally with total pressure.

38 [HARD] Equilibrium Principles & Quantitative Analysis 38 — 9701/13/M/J/223/Q33**[1]**

An industrial equilibrium chamber contains gas mixture 38. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [B] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [C] The equilibrium shifts to the side with more moles of gas.
- [D] The value of K_p increases proportionally with total pressure.

39 [EASY] Equilibrium Principles & Quantitative Analysis 39 — 9701/11/M/J/224/Q34**[1]**

An industrial equilibrium chamber contains gas mixture 39. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium shifts to the side with more moles of gas.
- [C] The value of K_p increases proportionally with total pressure.
- [D] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.

40 [HARD] Equilibrium Principles & Quantitative Analysis 40 — 9701/12/M/J/220/Q35**[1]**

An industrial equilibrium chamber contains gas mixture 40. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [C] The equilibrium shifts to the side with more moles of gas.
- [D] The value of K_p increases proportionally with total pressure.

41 [EASY] Equilibrium Principles & Quantitative Analysis 41 — 9701/13/M/J/221/Q36**[1]**

An industrial equilibrium chamber contains gas mixture 41. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [C] The equilibrium shifts to the side with more moles of gas.
- [D] The value of K_p increases proportionally with total pressure.

42 [HARD] Equilibrium Principles & Quantitative Analysis 42 — 9701/11/M/J/222/Q37**[1]**

An industrial equilibrium chamber contains gas mixture 42. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium shifts to the side with more moles of gas.
- [C] The value of K_p increases proportionally with total pressure.
- [D] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.

43 [EASY] Equilibrium Principles & Quantitative Analysis 43 — 9701/12/M/J/223/Q38**[1]**

An industrial equilibrium chamber contains gas mixture 43. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [B] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [C] The equilibrium shifts to the side with more moles of gas.
- [D] The value of K_p increases proportionally with total pressure.

44 [HARD] Equilibrium Principles & Quantitative Analysis 44 — 9701/13/M/J/224/Q39**[1]**

An industrial equilibrium chamber contains gas mixture 44. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium shifts to the side with more moles of gas.
- [C] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [D] The value of K_p increases proportionally with total pressure.

45 [EASY] Equilibrium Principles & Quantitative Analysis 45 — 9701/11/M/J/220/Q40**[1]**

An industrial equilibrium chamber contains gas mixture 45. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [B] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [C] The equilibrium shifts to the side with more moles of gas.
- [D] The value of K_p increases proportionally with total pressure.

46 [HARD] Equilibrium Principles & Quantitative Analysis 46 — 9701/12/M/J/221/Q41**[1]**

An industrial equilibrium chamber contains gas mixture 46. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium shifts to the side with more moles of gas.
- [C] The value of K_p increases proportionally with total pressure.
- [D] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.

47 [EASY] Equilibrium Principles & Quantitative Analysis 47 — 9701/13/M/J/222/Q42**[1]**

An industrial equilibrium chamber contains gas mixture 47. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [C] The equilibrium shifts to the side with more moles of gas.
- [D] The value of K_p increases proportionally with total pressure.

48 [HARD] Equilibrium Principles & Quantitative Analysis 48 — 9701/11/M/J/223/Q43**[1]**

An industrial equilibrium chamber contains gas mixture 48. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium shifts to the side with more moles of gas.
- [C] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [D] The value of K_p increases proportionally with total pressure.

49 [EASY] Equilibrium Principles & Quantitative Analysis 49 — 9701/12/M/J/224/Q44**[1]**

An industrial equilibrium chamber contains gas mixture 49. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [C] The equilibrium shifts to the side with more moles of gas.
- [D] The value of K_p increases proportionally with total pressure.

50 [HARD] Equilibrium Principles & Quantitative Analysis 50 — 9701/13/M/J/220/Q45**[1]**

An industrial equilibrium chamber contains gas mixture 50. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [B] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [C] The equilibrium shifts to the side with more moles of gas.
- [D] The value of K_p increases proportionally with total pressure.

51 [EASY] Equilibrium Principles & Quantitative Analysis 51 — 9701/11/M/J/221/Q46**[1]**

An industrial equilibrium chamber contains gas mixture 51. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium shifts to the side with more moles of gas.
- [C] The value of K_p increases proportionally with total pressure.
- [D] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.

52 [HARD] Equilibrium Principles & Quantitative Analysis 52 — 9701/12/M/J/222/Q47**[1]**

An industrial equilibrium chamber contains gas mixture 52. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium shifts to the side with more moles of gas.
- [C] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [D] The value of K_p increases proportionally with total pressure.

53 [EASY] Equilibrium Principles & Quantitative Analysis 53 — 9701/13/M/J/223/Q48**[1]**

An industrial equilibrium chamber contains gas mixture 53. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [B] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [C] The equilibrium shifts to the side with more moles of gas.
- [D] The value of K_p increases proportionally with total pressure.

54 [HARD] Equilibrium Principles & Quantitative Analysis 54 — 9701/11/M/J/224/Q49**[1]**

An industrial equilibrium chamber contains gas mixture 54. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium shifts to the side with more moles of gas.
- [C] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [D] The value of K_p increases proportionally with total pressure.

55 [EASY] Equilibrium Principles & Quantitative Analysis 55 — 9701/12/M/J/220/Q50**[1]**

An industrial equilibrium chamber contains gas mixture 55. If an inert gas such as argon is added at constant volume, what happens to the position of equilibrium?

- [A] The equilibrium shifts to the side with fewer moles of gas due to increased total pressure.
- [B] The equilibrium position remains unchanged because the partial pressures of the reacting gases are unaltered.
- [C] The equilibrium shifts to the side with more moles of gas.
- [D] The value of K_p increases proportionally with total pressure.

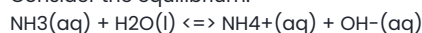
SUBTOPIC 7.2 — BRØNSTED-LOWRY ACIDS & BASES (Q56 – Q100)**56** [EASY] Brønsted-Lowry Definition of Acids and Bases — 9701/11/M/J/23/Q30**[1]**

According to the Brønsted-Lowry theory, how are an acid and a base defined?

- [A] An acid is an electron pair acceptor; a base is an electron pair donor.
- [B] An acid produces OH^- ions in water; a base produces H^+ ions in water.
- [C] An acid is a substance with $\text{pH} > 7$; a base is a substance with $\text{pH} < 7$.
- [D] An acid is a proton (H^+) donor; a base is a proton (H^+) acceptor.

57 [EASY] Conjugate Acid-Base Pairs in Aqueous Solutions — 9701/12/M/J/23/Q30**[1]**

Consider the equilibrium:



Which pair represents a conjugate acid-base pair in this system?

- [A] NH_3 (acid) and NH_4^+ (conjugate base)
- [B] H_2O (base) and NH_4^+ (conjugate acid)
- [C] H_2O (acid) and OH^- (conjugate base)
- [D] NH_3 (base) and OH^- (conjugate acid)

58 [EASY] Conjugate Base of the Hydrogensulfate Ion — 9701/13/M/J/23/Q30**[1]**

What is the conjugate base of the hydrogensulfate ion, HSO_4^- ?

- [A] SO_4^{2-}
- [B] H_2SO_4
- [C] H_3SO_4^+
- [D] SO_3^{2-}

59 [EASY] Conjugate Acid of the Monohydrogenphosphate Ion — 9701/11/O/N/23/Q32**[1]**

What is the conjugate acid of the monohydrogenphosphate ion, HPO_4^{2-} ?

- [A] PO_4^{3-}
- [B] H_3PO_4
- [C] $\text{P}_2\text{O}_7^{4-}$
- [D] H_2PO_4^-

60 [HARD] Strong vs Weak Acids: Physical and Chemical Properties — 9701/12/O/N/23/Q32**[1]**

A student prepares equimolar (0.10 mol dm^{-3}) solutions of hydrochloric acid, $\text{HCl}(\text{aq})$, and ethanoic acid, $\text{CH}_3\text{COOH}(\text{aq})$.

Which observation is identical for both solutions?

- [A]** The electrical conductivity of the two solutions.
- [B]** The volume of 0.10 mol dm^{-3} NaOH required for complete neutralisation of 25.0 cm^3 of each acid.
- [C]** The pH of the two solutions measured with a pH meter.
- [D]** The initial rate of effervescence when magnesium ribbon is added.

61 [HARD] Acid-Base Equilibria & Proton Transfer 61 — 9701/12/O/N/221/Q26**[1]**

In a non-aqueous acid-base system 61, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A]** Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B]** Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [C]** Lewis acid, because it donates an electron pair.
- [D]** Reducing agent, because nitrogen is reduced.

62 [EASY] Acid-Base Equilibria & Proton Transfer 62 — 9701/13/O/N/222/Q27**[1]**

In a non-aqueous acid-base system 62, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A]** Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B]** Lewis acid, because it donates an electron pair.
- [C]** Reducing agent, because nitrogen is reduced.
- [D]** Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .

63 [HARD] Acid-Base Equilibria & Proton Transfer 63 — 9701/11/O/N/223/Q28**[1]**

In a non-aqueous acid-base system 63, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A]** Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [B]** Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [C]** Lewis acid, because it donates an electron pair.
- [D]** Reducing agent, because nitrogen is reduced.

64 [EASY] Acid-Base Equilibria & Proton Transfer 64 — 9701/12/O/N/224/Q29**[1]**

In a non-aqueous acid-base system 64, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A]** Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B]** Lewis acid, because it donates an electron pair.
- [C]** Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [D]** Reducing agent, because nitrogen is reduced.

65 [HARD] Acid-Base Equilibria & Proton Transfer 65 — 9701/13/O/N/220/Q30**[1]**

In a non-aqueous acid-base system 65, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [B] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [C] Lewis acid, because it donates an electron pair.
- [D] Reducing agent, because nitrogen is reduced.

66 [EASY] Acid-Base Equilibria & Proton Transfer 66 — 9701/11/O/N/221/Q31**[1]**

In a non-aqueous acid-base system 66, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B] Lewis acid, because it donates an electron pair.
- [C] Reducing agent, because nitrogen is reduced.
- [D] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .

67 [HARD] Acid-Base Equilibria & Proton Transfer 67 — 9701/12/O/N/222/Q32**[1]**

In a non-aqueous acid-base system 67, pure liquid nitric acid reacts with pure sulfuric acid according to:

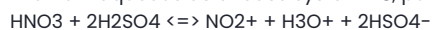


What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [C] Lewis acid, because it donates an electron pair.
- [D] Reducing agent, because nitrogen is reduced.

68 [EASY] Acid-Base Equilibria & Proton Transfer 68 — 9701/13/O/N/223/Q33**[1]**

In a non-aqueous acid-base system 68, pure liquid nitric acid reacts with pure sulfuric acid according to:

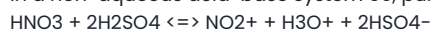


What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B] Lewis acid, because it donates an electron pair.
- [C] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [D] Reducing agent, because nitrogen is reduced.

69 [HARD] Acid-Base Equilibria & Proton Transfer 69 — 9701/11/O/N/224/Q34**[1]**

In a non-aqueous acid-base system 69, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [C] Lewis acid, because it donates an electron pair.
- [D] Reducing agent, because nitrogen is reduced.

70 [EASY] Acid-Base Equilibria & Proton Transfer 70 — 9701/12/O/N/220/Q35

[1]

In a non-aqueous acid-base system 70, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [B] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [C] Lewis acid, because it donates an electron pair.
- [D] Reducing agent, because nitrogen is reduced.

71 [HARD] Acid-Base Equilibria & Proton Transfer 71 — 9701/13/O/N/221/Q36

[1]

In a non-aqueous acid-base system 71, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B] Lewis acid, because it donates an electron pair.
- [C] Reducing agent, because nitrogen is reduced.
- [D] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .

72 [EASY] Acid-Base Equilibria & Proton Transfer 72 — 9701/11/O/N/222/Q37

[1]

In a non-aqueous acid-base system 72, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B] Lewis acid, because it donates an electron pair.
- [C] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [D] Reducing agent, because nitrogen is reduced.

73 [HARD] Acid-Base Equilibria & Proton Transfer 73 — 9701/12/O/N/223/Q38

[1]

In a non-aqueous acid-base system 73, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [B] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [C] Lewis acid, because it donates an electron pair.
- [D] Reducing agent, because nitrogen is reduced.

74 [EASY] Acid-Base Equilibria & Proton Transfer 74 — 9701/13/O/N/224/Q39

[1]

In a non-aqueous acid-base system 74, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B] Lewis acid, because it donates an electron pair.
- [C] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [D] Reducing agent, because nitrogen is reduced.

75 [HARD] Acid-Base Equilibria & Proton Transfer 75 — 9701/11/O/N/220/Q40**[1]**

In a non-aqueous acid-base system 75, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [C] Lewis acid, because it donates an electron pair.
- [D] Reducing agent, because nitrogen is reduced.

76 [EASY] Acid-Base Equilibria & Proton Transfer 76 — 9701/12/O/N/221/Q41**[1]**

In a non-aqueous acid-base system 76, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B] Lewis acid, because it donates an electron pair.
- [C] Reducing agent, because nitrogen is reduced.
- [D] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .

77 [HARD] Acid-Base Equilibria & Proton Transfer 77 — 9701/13/O/N/222/Q42**[1]**

In a non-aqueous acid-base system 77, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B] Lewis acid, because it donates an electron pair.
- [C] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [D] Reducing agent, because nitrogen is reduced.

78 [EASY] Acid-Base Equilibria & Proton Transfer 78 — 9701/11/O/N/223/Q43**[1]**

In a non-aqueous acid-base system 78, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [B] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [C] Lewis acid, because it donates an electron pair.
- [D] Reducing agent, because nitrogen is reduced.

79 [HARD] Acid-Base Equilibria & Proton Transfer 79 — 9701/12/O/N/224/Q44**[1]**

In a non-aqueous acid-base system 79, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B] Lewis acid, because it donates an electron pair.
- [C] Reducing agent, because nitrogen is reduced.
- [D] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .

80 [EASY] Acid-Base Equilibria & Proton Transfer 80 — 9701/13/O/N/220/Q45**[1]**

In a non-aqueous acid-base system 80, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [C] Lewis acid, because it donates an electron pair.
- [D] Reducing agent, because nitrogen is reduced.

81 [HARD] Acid-Base Equilibria & Proton Transfer 81 — 9701/11/O/N/221/Q46**[1]**

In a non-aqueous acid-base system 81, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [C] Lewis acid, because it donates an electron pair.
- [D] Reducing agent, because nitrogen is reduced.

82 [EASY] Acid-Base Equilibria & Proton Transfer 82 — 9701/12/O/N/222/Q47**[1]**

In a non-aqueous acid-base system 82, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B] Lewis acid, because it donates an electron pair.
- [C] Reducing agent, because nitrogen is reduced.
- [D] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .

83 [HARD] Acid-Base Equilibria & Proton Transfer 83 — 9701/13/O/N/223/Q48**[1]**

In a non-aqueous acid-base system 83, pure liquid nitric acid reacts with pure sulfuric acid according to:

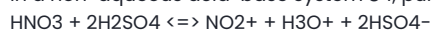


What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [B] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [C] Lewis acid, because it donates an electron pair.
- [D] Reducing agent, because nitrogen is reduced.

84 [EASY] Acid-Base Equilibria & Proton Transfer 84 — 9701/11/O/N/224/Q49**[1]**

In a non-aqueous acid-base system 84, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B] Lewis acid, because it donates an electron pair.
- [C] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [D] Reducing agent, because nitrogen is reduced.

85 [HARD] Acid-Base Equilibria & Proton Transfer 85 — 9701/12/O/N/220/Q50**[1]**

In a non-aqueous acid-base system 85, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [B] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [C] Lewis acid, because it donates an electron pair.
- [D] Reducing agent, because nitrogen is reduced.

86 [EASY] Acid-Base Equilibria & Proton Transfer 86 — 9701/13/O/N/221/Q51**[1]**

In a non-aqueous acid-base system 86, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B] Lewis acid, because it donates an electron pair.
- [C] Reducing agent, because nitrogen is reduced.
- [D] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .

87 [HARD] Acid-Base Equilibria & Proton Transfer 87 — 9701/11/O/N/222/Q52**[1]**

In a non-aqueous acid-base system 87, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [C] Lewis acid, because it donates an electron pair.
- [D] Reducing agent, because nitrogen is reduced.

88 [EASY] Acid-Base Equilibria & Proton Transfer 88 — 9701/12/O/N/223/Q53**[1]**

In a non-aqueous acid-base system 88, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B] Lewis acid, because it donates an electron pair.
- [C] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [D] Reducing agent, because nitrogen is reduced.

89 [HARD] Acid-Base Equilibria & Proton Transfer 89 — 9701/13/O/N/224/Q54**[1]**

In a non-aqueous acid-base system 89, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [C] Lewis acid, because it donates an electron pair.
- [D] Reducing agent, because nitrogen is reduced.

90 [EASY] Acid-Base Equilibria & Proton Transfer 90 — 9701/11/O/N/220/Q55**[1]**

In a non-aqueous acid-base system 90, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [B] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [C] Lewis acid, because it donates an electron pair.
- [D] Reducing agent, because nitrogen is reduced.

91 [HARD] Acid-Base Equilibria & Proton Transfer 91 — 9701/12/O/N/221/Q56**[1]**

In a non-aqueous acid-base system 91, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B] Lewis acid, because it donates an electron pair.
- [C] Reducing agent, because nitrogen is reduced.
- [D] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .

92 [EASY] Acid-Base Equilibria & Proton Transfer 92 — 9701/13/O/N/222/Q57**[1]**

In a non-aqueous acid-base system 92, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B] Lewis acid, because it donates an electron pair.
- [C] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [D] Reducing agent, because nitrogen is reduced.

93 [HARD] Acid-Base Equilibria & Proton Transfer 93 — 9701/11/O/N/223/Q58**[1]**

In a non-aqueous acid-base system 93, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [B] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [C] Lewis acid, because it donates an electron pair.
- [D] Reducing agent, because nitrogen is reduced.

94 [EASY] Acid-Base Equilibria & Proton Transfer 94 — 9701/12/O/N/224/Q59**[1]**

In a non-aqueous acid-base system 94, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B] Lewis acid, because it donates an electron pair.
- [C] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [D] Reducing agent, because nitrogen is reduced.

95 [HARD] Acid-Base Equilibria & Proton Transfer 95 — 9701/13/O/N/220/Q60**[1]**

In a non-aqueous acid-base system 95, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [C] Lewis acid, because it donates an electron pair.
- [D] Reducing agent, because nitrogen is reduced.

96 [EASY] Acid-Base Equilibria & Proton Transfer 96 — 9701/11/O/N/221/Q61**[1]**

In a non-aqueous acid-base system 96, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B] Lewis acid, because it donates an electron pair.
- [C] Reducing agent, because nitrogen is reduced.
- [D] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .

97 [HARD] Acid-Base Equilibria & Proton Transfer 97 — 9701/12/O/N/222/Q62**[1]**

In a non-aqueous acid-base system 97, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B] Lewis acid, because it donates an electron pair.
- [C] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [D] Reducing agent, because nitrogen is reduced.

98 [EASY] Acid-Base Equilibria & Proton Transfer 98 — 9701/13/O/N/223/Q63**[1]**

In a non-aqueous acid-base system 98, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [B] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [C] Lewis acid, because it donates an electron pair.
- [D] Reducing agent, because nitrogen is reduced.

99 [HARD] Acid-Base Equilibria & Proton Transfer 99 — 9701/11/O/N/224/Q64**[1]**

In a non-aqueous acid-base system 99, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B] Lewis acid, because it donates an electron pair.
- [C] Reducing agent, because nitrogen is reduced.
- [D] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .

100 [EASY] Acid-Base Equilibria & Proton Transfer 100 — 9701/12/O/N/220/Q65**[1]**

In a non-aqueous acid-base system 100, pure liquid nitric acid reacts with pure sulfuric acid according to:



What role does HNO_3 play in this reaction?

- [A] Brønsted-Lowry acid, because it donates a proton to form HSO_4^- .
- [B] Brønsted-Lowry base, because it accepts a proton from the stronger acid H_2SO_4 .
- [C] Lewis acid, because it donates an electron pair.
- [D] Reducing agent, because nitrogen is reduced.

FREQUENTLY EXAMINED CORE QUESTIONS — HIGH-FREQUENCY REPEATS (Q101 – Q110)**101** [EASY] [HIGH-FREQUENCY EXAM REPEAT] Effect of Catalyst on Chemical Equilibrium Position — 9701/11/M/J/23/Q31**[1]**

Which statement concerning the effect of a catalyst on a dynamic reversible reaction is correct?

- [A] It increases the yield of products at equilibrium by lowering activation energy.
- [B] It increases the equilibrium constant K_c for endothermic reactions only.
- [C] It increases the rate of both forward and reverse reactions equally, establishing equilibrium faster without altering the equilibrium position or the value of K_c .
- [D] It alters the enthalpy change ΔH of the reaction.

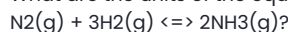
102 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Temperature Dependence of the Equilibrium Constant — 9701/12/M/J/23/Q31**[1]**

For an endothermic reversible reaction ($\Delta H > 0$), what happens to the value of K_c when the temperature is raised?

- [A] K_c increases because the equilibrium shifts to the right, favoring product formation.
- [B] K_c decreases because higher temperature favors the reverse reaction.
- [C] K_c remains constant because it is an equilibrium constant.
- [D] K_c changes only if the pressure is also altered.

103 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Deducing Units of K_c for Haber Process — 9701/13/M/J/23/Q31**[1]**

What are the units of the equilibrium constant K_c for the reaction:



- [A] $\text{mol}^2 \text{dm}^{-6}$
- [B] $\text{mol}^{-1} \text{dm}^3$
- [C] No units
- [D] $\text{mol}^{-2} \text{dm}^6$

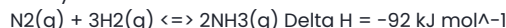
104 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Identifying Conjugate Acid-Base Pairs — 9701/11/O/N/23/Q33**[1]**

What is the conjugate base of the dihydrogenphosphate ion, H_2PO_4^- ?

- [A] H_3PO_4
- [B] HPO_4^{2-}
- [C] PO_4^{3-}
- [D] H_4PO_4^+

105 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Compromise Conditions in the Industrial Haber Process — 9701/12/O/N/23/Q33**[1]**

The synthesis of ammonia is exothermic:



Why is an operating temperature of 400 – 450 °C utilized industrially rather than room temperature?

- [A]** 400 – 450 °C is an economic compromise between high equilibrium yield (favored at low temperatures) and an acceptable reaction rate (favored at high temperatures).
- [B]** Ammonia decomposes spontaneously at temperatures below 400 °C.
- [C]** The iron catalyst is completely inactive at temperatures above 300 °C.
- [D]** High temperatures shift the equilibrium position to the right, maximizing conversion.

106 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Calculation of K_c from Equilibrium Concentrations — 9701/13/O/N/23/Q33**[1]**

In a 2.0 dm³ sealed vessel, an equilibrium mixture contains 0.40 mol of SO₂, 0.20 mol of O₂, and 0.80 mol of SO₃ at temperature T:

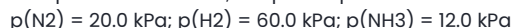
$$2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$$

What is the value of the equilibrium constant K_c at this temperature?

- [A]** 20 dm³ mol⁻¹
- [B]** 8.0 dm³ mol⁻¹
- [C]** 40 dm³ mol⁻¹
- [D]** 80 dm³ mol⁻¹

107 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Calculation of K_p from Equilibrium Partial Pressures — 9701/11/F/M/24/Q25**[1]**

At equilibrium at 500 K, the partial pressures in the Haber synthesis are:



What is the numerical value of K_p for $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$?

- [A]** $1.00 \times 10^{-4} \text{ kPa}^{-2}$
- [B]** $3.33 \times 10^{-5} \text{ kPa}^{-2}$
- [C]** $2.50 \times 10^{-3} \text{ kPa}^{-2}$
- [D]** $6.67 \times 10^{-5} \text{ kPa}^{-2}$

108 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Reversible Gas Reaction with No Pressure Effect — 9701/12/F/M/24/Q25**[1]**

For the equilibrium: $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$, what effect does doubling the total pressure at constant temperature have on the equilibrium yield of HI?

- [A]** The yield doubles because pressure increases collision frequency.
- [B]** The yield is halved because HI molecules decompose under pressure.
- [C]** The value of K_p increases fourfold.
- [D]** No effect, because the number of gaseous moles on both sides of the equation is equal.

109 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Equimolar Strong vs Weak Acid Neutralisation Titre — 9701/13/F/M/24/Q25**[1]**

Sample A contains 25.0 cm³ of 0.10 mol dm⁻³ HCl(aq).

Sample B contains 25.0 cm³ of 0.10 mol dm⁻³ CH₃COOH(aq).

Both samples are titrated against 0.10 mol dm⁻³ NaOH(aq).

Which statement is correct?

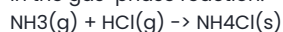
- [A]** Both samples require exactly the same volume (25.0 cm³) of NaOH(aq) for complete neutralisation.
- [B]** Sample A requires more NaOH because HCl is a strong acid.
- [C]** Sample B requires more NaOH because ethanoic acid is a weak acid.
- [D]** Sample B neutralises faster than Sample A.

T10 [HARD] [HIGH-FREQUENCY EXAM REPEAT]

Non-Aqueous Proton Transfer and Brønsted-Lowry Theory — 9701/11/M/J/22/Q30

[1]

In the gas-phase reaction:



How is this reaction classified according to the Brønsted-Lowry theory?

- [A] A redox reaction where chlorine is reduced.
- [B] A precipitation reaction involving spectator ions.
- [C] An acid-base reaction where HCl donates a proton (acid) to NH_3 (base).
- [D] A free-radical addition reaction.

QUICK-CHECK ANSWER KEY GRID (Q1 – Q110)

Official Cambridge 9701 Paper 1 Marking Matrix · 1 Mark Per Question · Total Marks: 110

Q1 – Q20		Q21 – Q40		Q41 – Q60		Q61 – Q80		Q81 – Q100	
Q01	B	Q21	B	Q41	B	Q61	B	Q81	B
Q02	D	Q22	D	Q42	D	Q62	D	Q82	D
Q03	A	Q23	A	Q43	A	Q63	A	Q83	A
Q04	C	Q24	C	Q44	C	Q64	C	Q84	C
Q05	A	Q25	A	Q45	A	Q65	A	Q85	A
Q06	D	Q26	D	Q46	D	Q66	D	Q86	D
Q07	B	Q27	B	Q47	B	Q67	B	Q87	B
Q08	C	Q28	C	Q48	C	Q68	C	Q88	C
Q09	B	Q29	B	Q49	B	Q69	B	Q89	B
Q10	A	Q30	A	Q50	A	Q70	A	Q90	A
Q11	D	Q31	D	Q51	D	Q71	D	Q91	D
Q12	C	Q32	C	Q52	C	Q72	C	Q92	C
Q13	A	Q33	A	Q53	A	Q73	A	Q93	A
Q14	C	Q34	C	Q54	C	Q74	C	Q94	C
Q15	B	Q35	B	Q55	B	Q75	B	Q95	B
Q16	D	Q36	D	Q56	D	Q76	D	Q96	D
Q17	C	Q37	C	Q57	C	Q77	C	Q97	C
Q18	A	Q38	A	Q58	A	Q78	A	Q98	A
Q19	D	Q39	D	Q59	D	Q79	D	Q99	D
Q20	B	Q40	B	Q60	B	Q80	B	Q100	B

FREQUENTLY EXAMINED CORE QUESTIONS (Q101 – Q110)

Q101 C	Q102 A	Q103 D	Q104 B	Q105 A	Q106 C	Q107 B	Q108 D	Q109 A	Q110 C
-----------	-----------	-----------	-----------	-----------	-----------	-----------	-----------	-----------	-----------

COMPREHENSIVE EXAMINER EXPLANATIONS & DISTRACTOR ANALYSIS

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
1	B	Option B is the precise definition of dynamic equilibrium: the forward and reverse reactions proceed simultaneously at identical rates (dynamic), with no net macroscopic change in the concentrations or physical properties of reactants and products over time.	[1]
2	D	Option D is correct. A catalyst provides an alternative reaction pathway with a lower activation energy. Because it lowers the activation energy barrier by the identical amount for both the forward and reverse directions, it accelerates both reactions equally. Consequently, the equilibrium composition and the numerical value of the equilibrium constant K_c remain completely unchanged.	[1]
3	A	Option A is correct. The equilibrium constant K_c is a true thermodynamic constant whose value depends SOLELY on temperature. Changes in pressure, concentration, or the addition of a catalyst will shift the equilibrium position (adjusting individual equilibrium concentrations) in accordance with Le Chatelier's principle, but the ratio defined by K_c remains strictly constant unless temperature is changed.	[1]
4	C	Option C is correct. According to Le Chatelier's principle, increasing temperature favors the endothermic direction (the reverse reaction, absorbing heat). Consequently, the equilibrium shifts to the left, decreasing the partial pressure of NH_3 and increasing the partial pressures of N_2 and H_2 . Because product partial pressures decrease while reactant partial pressures increase, the value of $K_p = p(\text{NH}_3)^2 / [p(\text{N}_2) \times p(\text{H}_2)^3]$ strictly decreases.	[1]
5	A	Option A is correct. On the left side, there are $2 + 1 = 3$ moles of gas; on the right side, there are 2 moles of gas. Increasing pressure causes the system to respond by shifting in the direction that produces fewer moles of gas (to the right), reducing the total number of gas molecules and mitigating the pressure rise. The value of K_p remains constant.	[1]
6	D	Option D is correct. The expression for K_c is: $K_c = [\text{CH}_3\text{COOC}_2\text{H}_5][\text{H}_2\text{O}] / ([\text{CH}_3\text{COOH}][\text{C}_2\text{H}_5\text{OH}])$. Units = $(\text{mol dm}^{-3} \times \text{mol dm}^{-3}) / (\text{mol dm}^{-3} \times \text{mol dm}^{-3})$. All concentration units cancel out completely, leaving K_c dimensionless (no units).	[1]
7	B	Option B is correct. $K_p = p(\text{NH}_3)^2 / [p(\text{N}_2) \times p(\text{H}_2)^3]$. Units = $(\text{kPa})^2 / [\text{kPa} \times (\text{kPa})^3] = \text{kPa}^2 / \text{kPa}^4 = \text{kPa}^{-2}$.	[1]
8	C	Option C is correct. Total moles = $2.0 + 1.0 + 7.0 = 10.0$ mol. Mole fraction of SO_2 : $x(\text{SO}_2) = 2.0 / 10.0 = 0.20$. By Dalton's law: $p(\text{SO}_2) = x(\text{SO}_2) \times p_{\text{total}} = 0.20 \times 400 \text{ kPa} = 80 \text{ kPa}$.	[1]
9	B	Option B is correct. In $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$, there are exactly 2 moles of gas on the reactant side ($1 \text{ H}_2 + 1 \text{ I}_2$) and 2 moles of gas on the product side (2 HI). Because the total number of gas molecules is identical on both sides of the equation ($\Delta n_{\text{gas}} = 0$), a change in volume or pressure affects both directions equally, causing no shift in equilibrium position.	[1]
10	A	Option A is correct. Initial moles: ester = 1.00, water = 1.00, acid = 0, alcohol = 0. Change: -0.33, -0.33, +0.33, +0.33. Equilibrium moles: ester = 0.67, water = 0.67, acid = 0.33, alcohol = 0.33. Since volume is 1.0 dm ³ , moles equal concentrations. $K_c = [\text{acid}][\text{alcohol}] / ([\text{ester}][\text{water}]) = (0.33 \times 0.33) / (0.67 \times 0.67) = 0.1089 / 0.4489 = 0.243 \rightarrow 0.24$.	[1]
11	D	Option D is correct. Adding an inert gas at constant volume increases the total pressure, but does NOT change the volume of the container or the concentration (or partial pressure) of any reacting species ($p_i = n_i RT / V$ remains constant). Because individual partial pressures are unchanged, the reaction quotient Q remains equal to K_p , and the equilibrium position does not shift.	[1]
12	C	Option C is correct. Adding an inert gas at constant volume increases the total pressure, but does NOT change the volume of the container or the concentration (or partial pressure) of any reacting species ($p_i = n_i RT / V$ remains constant). Because individual partial pressures are unchanged, the reaction quotient Q remains equal to K_p , and the equilibrium position does not shift.	[1]
13	A	Option A is correct. Adding an inert gas at constant volume increases the total pressure, but does NOT change the volume of the container or the concentration (or partial pressure) of any reacting species ($p_i = n_i RT / V$ remains constant). Because individual partial pressures are unchanged, the reaction quotient Q remains equal to K_p , and the equilibrium position does not shift.	[1]
14	C	Option C is correct. Adding an inert gas at constant volume increases the total pressure, but does NOT change the volume of the container or the concentration (or partial pressure) of any reacting species ($p_i = n_i RT / V$ remains constant). Because individual partial pressures are unchanged, the reaction quotient Q remains equal to K_p , and the equilibrium position does not shift.	[1]
15	B	Option B is correct. Adding an inert gas at constant volume increases the total pressure, but does NOT change the volume of the container or the concentration (or partial pressure) of any reacting species ($p_i = n_i RT / V$ remains constant). Because individual partial pressures are unchanged, the reaction quotient Q remains equal to K_p , and the equilibrium position does not shift.	[1]
16	D	Option D is correct. Adding an inert gas at constant volume increases the total pressure, but does NOT change the volume of the container or the concentration (or partial pressure) of any reacting species ($p_i = n_i RT / V$ remains constant). Because individual partial pressures are unchanged, the reaction quotient Q remains equal to K_p , and the equilibrium position does not shift.	[1]
17	C	Option C is correct. Adding an inert gas at constant volume increases the total pressure, but does NOT change the volume of the container or the concentration (or partial pressure) of any reacting species ($p_i = n_i RT / V$ remains constant). Because individual partial pressures are unchanged, the reaction quotient Q remains equal to K_p , and the equilibrium position does not shift.	[1]

25

26

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
48	C	Option C is correct. Adding an inert gas at constant volume increases the total pressure, but does NOT change the volume of the container or the concentration (or partial pressure) of any reacting species ($p_i = n_i RT / V$ remains constant). Because individual partial pressures are unchanged, the reaction quotient Q remains equal to K_p , and the equilibrium position does not shift.	[1]
49	B	Option B is correct. Adding an inert gas at constant volume increases the total pressure, but does NOT change the volume of the container or the concentration (or partial pressure) of any reacting species ($p_i = n_i RT / V$ remains constant). Because individual partial pressures are unchanged, the reaction quotient Q remains equal to K_p , and the equilibrium position does not shift.	[1]
50	A	Option A is correct. Adding an inert gas at constant volume increases the total pressure, but does NOT change the volume of the container or the concentration (or partial pressure) of any reacting species ($p_i = n_i RT / V$ remains constant). Because individual partial pressures are unchanged, the reaction quotient Q remains equal to K_p , and the equilibrium position does not shift.	[1]
51	D	Option D is correct. Adding an inert gas at constant volume increases the total pressure, but does NOT change the volume of the container or the concentration (or partial pressure) of any reacting species ($p_i = n_i RT / V$ remains constant). Because individual partial pressures are unchanged, the reaction quotient Q remains equal to K_p , and the equilibrium position does not shift.	[1]
52	C	Option C is correct. Adding an inert gas at constant volume increases the total pressure, but does NOT change the volume of the container or the concentration (or partial pressure) of any reacting species ($p_i = n_i RT / V$ remains constant). Because individual partial pressures are unchanged, the reaction quotient Q remains equal to K_p , and the equilibrium position does not shift.	[1]
53	A	Option A is correct. Adding an inert gas at constant volume increases the total pressure, but does NOT change the volume of the container or the concentration (or partial pressure) of any reacting species ($p_i = n_i RT / V$ remains constant). Because individual partial pressures are unchanged, the reaction quotient Q remains equal to K_p , and the equilibrium position does not shift.	[1]
54	C	Option C is correct. Adding an inert gas at constant volume increases the total pressure, but does NOT change the volume of the container or the concentration (or partial pressure) of any reacting species ($p_i = n_i RT / V$ remains constant). Because individual partial pressures are unchanged, the reaction quotient Q remains equal to K_p , and the equilibrium position does not shift.	[1]
55	B	Option B is correct. Adding an inert gas at constant volume increases the total pressure, but does NOT change the volume of the container or the concentration (or partial pressure) of any reacting species ($p_i = n_i RT / V$ remains constant). Because individual partial pressures are unchanged, the reaction quotient Q remains equal to K_p , and the equilibrium position does not shift.	[1]
56	D	Option D is the formal Brønsted-Lowry definition. (Option A describes the Lewis acid-base definition).	[1]
57	C	Option C is correct. A conjugate acid-base pair consists of two species that differ by exactly one proton (H^+). H_2O donates a proton to become OH^- , making H_2O an acid and OH^- its conjugate base. Likewise, NH_3 accepts a proton to become NH_4^+ , making NH_3 a base and NH_4^+ its conjugate acid.	[1]
58	A	Option A is correct. To find the conjugate base of any species, remove one proton (H^+): HSO_4^- minus H^+ yields the sulfate ion, SO_4^{2-} .	[1]
59	D	Option D is correct. To find the conjugate acid of any species, add one proton (H^+): HPO_4^{2-} plus H^+ yields the dihydrogenphosphate ion, $H_2PO_4^-$.	[1]
60	B	Option B is correct. Both HCl and CH_3COOH are monoprotic acids containing identical total moles of titratable acid in 25.0 cm^3 (0.0025 mol). Therefore, both require exactly the same volume of $0.10\text{ mol dm}^{-3}\text{ NaOH}$ for neutralisation (25.0 cm^3). However, because HCl is 100% dissociated while CH_3COOH is <2% dissociated, HCl has a much higher $[H^+]$, resulting in lower pH, higher electrical conductivity, and a much faster initial rate of reaction with magnesium.	[1]
61	B	Option B is correct. Sulfuric acid is a stronger acid than nitric acid. H_2SO_4 protonates HNO_3 to form $[H_2NO_3]^+$, which subsequently loses water to yield the nitronium ion, NO_2^+ . Because HNO_3 accepts a proton from H_2SO_4 , it behaves as a Brønsted-Lowry base in this medium.	[1]
62	D	Option D is correct. Sulfuric acid is a stronger acid than nitric acid. H_2SO_4 protonates HNO_3 to form $[H_2NO_3]^+$, which subsequently loses water to yield the nitronium ion, NO_2^+ . Because HNO_3 accepts a proton from H_2SO_4 , it behaves as a Brønsted-Lowry base in this medium.	[1]
63	A	Option A is correct. Sulfuric acid is a stronger acid than nitric acid. H_2SO_4 protonates HNO_3 to form $[H_2NO_3]^+$, which subsequently loses water to yield the nitronium ion, NO_2^+ . Because HNO_3 accepts a proton from H_2SO_4 , it behaves as a Brønsted-Lowry base in this medium.	[1]
64	C	Option C is correct. Sulfuric acid is a stronger acid than nitric acid. H_2SO_4 protonates HNO_3 to form $[H_2NO_3]^+$, which subsequently loses water to yield the nitronium ion, NO_2^+ . Because HNO_3 accepts a proton from H_2SO_4 , it behaves as a Brønsted-Lowry base in this medium.	[1]
65	A	Option A is correct. Sulfuric acid is a stronger acid than nitric acid. H_2SO_4 protonates HNO_3 to form $[H_2NO_3]^+$, which subsequently loses water to yield the nitronium ion, NO_2^+ . Because HNO_3 accepts a proton from H_2SO_4 , it behaves as a Brønsted-Lowry base in this medium.	[1]

[illegible]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
86	D	Option D is correct. Sulfuric acid is a stronger acid than nitric acid. H_2SO_4 protonates HNO_3 to form $[\text{H}_2\text{NO}_3]^+$, which subsequently loses water to yield the nitronium ion, NO_2^+ . Because HNO_3 accepts a proton from H_2SO_4 , it behaves as a Brønsted-Lowry base in this medium.	[1]
87	B	Option B is correct. Sulfuric acid is a stronger acid than nitric acid. H_2SO_4 protonates HNO_3 to form $[\text{H}_2\text{NO}_3]^+$, which subsequently loses water to yield the nitronium ion, NO_2^+ . Because HNO_3 accepts a proton from H_2SO_4 , it behaves as a Brønsted-Lowry base in this medium.	[1]
88	C	Option C is correct. Sulfuric acid is a stronger acid than nitric acid. H_2SO_4 protonates HNO_3 to form $[\text{H}_2\text{NO}_3]^+$, which subsequently loses water to yield the nitronium ion, NO_2^+ . Because HNO_3 accepts a proton from H_2SO_4 , it behaves as a Brønsted-Lowry base in this medium.	[1]
89	B	Option B is correct. Sulfuric acid is a stronger acid than nitric acid. H_2SO_4 protonates HNO_3 to form $[\text{H}_2\text{NO}_3]^+$, which subsequently loses water to yield the nitronium ion, NO_2^+ . Because HNO_3 accepts a proton from H_2SO_4 , it behaves as a Brønsted-Lowry base in this medium.	[1]
90	A	Option A is correct. Sulfuric acid is a stronger acid than nitric acid. H_2SO_4 protonates HNO_3 to form $[\text{H}_2\text{NO}_3]^+$, which subsequently loses water to yield the nitronium ion, NO_2^+ . Because HNO_3 accepts a proton from H_2SO_4 , it behaves as a Brønsted-Lowry base in this medium.	[1]
91	D	Option D is correct. Sulfuric acid is a stronger acid than nitric acid. H_2SO_4 protonates HNO_3 to form $[\text{H}_2\text{NO}_3]^+$, which subsequently loses water to yield the nitronium ion, NO_2^+ . Because HNO_3 accepts a proton from H_2SO_4 , it behaves as a Brønsted-Lowry base in this medium.	[1]
92	C	Option C is correct. Sulfuric acid is a stronger acid than nitric acid. H_2SO_4 protonates HNO_3 to form $[\text{H}_2\text{NO}_3]^+$, which subsequently loses water to yield the nitronium ion, NO_2^+ . Because HNO_3 accepts a proton from H_2SO_4 , it behaves as a Brønsted-Lowry base in this medium.	[1]
93	A	Option A is correct. Sulfuric acid is a stronger acid than nitric acid. H_2SO_4 protonates HNO_3 to form $[\text{H}_2\text{NO}_3]^+$, which subsequently loses water to yield the nitronium ion, NO_2^+ . Because HNO_3 accepts a proton from H_2SO_4 , it behaves as a Brønsted-Lowry base in this medium.	[1]
94	C	Option C is correct. Sulfuric acid is a stronger acid than nitric acid. H_2SO_4 protonates HNO_3 to form $[\text{H}_2\text{NO}_3]^+$, which subsequently loses water to yield the nitronium ion, NO_2^+ . Because HNO_3 accepts a proton from H_2SO_4 , it behaves as a Brønsted-Lowry base in this medium.	[1]
95	B	Option B is correct. Sulfuric acid is a stronger acid than nitric acid. H_2SO_4 protonates HNO_3 to form $[\text{H}_2\text{NO}_3]^+$, which subsequently loses water to yield the nitronium ion, NO_2^+ . Because HNO_3 accepts a proton from H_2SO_4 , it behaves as a Brønsted-Lowry base in this medium.	[1]
96	D	Option D is correct. Sulfuric acid is a stronger acid than nitric acid. H_2SO_4 protonates HNO_3 to form $[\text{H}_2\text{NO}_3]^+$, which subsequently loses water to yield the nitronium ion, NO_2^+ . Because HNO_3 accepts a proton from H_2SO_4 , it behaves as a Brønsted-Lowry base in this medium.	[1]
97	C	Option C is correct. Sulfuric acid is a stronger acid than nitric acid. H_2SO_4 protonates HNO_3 to form $[\text{H}_2\text{NO}_3]^+$, which subsequently loses water to yield the nitronium ion, NO_2^+ . Because HNO_3 accepts a proton from H_2SO_4 , it behaves as a Brønsted-Lowry base in this medium.	[1]
98	A	Option A is correct. Sulfuric acid is a stronger acid than nitric acid. H_2SO_4 protonates HNO_3 to form $[\text{H}_2\text{NO}_3]^+$, which subsequently loses water to yield the nitronium ion, NO_2^+ . Because HNO_3 accepts a proton from H_2SO_4 , it behaves as a Brønsted-Lowry base in this medium.	[1]
99	D	Option D is correct. Sulfuric acid is a stronger acid than nitric acid. H_2SO_4 protonates HNO_3 to form $[\text{H}_2\text{NO}_3]^+$, which subsequently loses water to yield the nitronium ion, NO_2^+ . Because HNO_3 accepts a proton from H_2SO_4 , it behaves as a Brønsted-Lowry base in this medium.	[1]
100	B	Option B is correct. Sulfuric acid is a stronger acid than nitric acid. H_2SO_4 protonates HNO_3 to form $[\text{H}_2\text{NO}_3]^+$, which subsequently loses water to yield the nitronium ion, NO_2^+ . Because HNO_3 accepts a proton from H_2SO_4 , it behaves as a Brønsted-Lowry base in this medium.	[1]
101	C	Option C is correct. A catalyst increases the rates of both forward and reverse reactions by providing an alternative mechanism with a lower activation energy barrier. Because the barrier is lowered equally in both directions, the position of equilibrium and the numerical value of the equilibrium constant K_c remain completely unchanged.	[1]
102	A	Option A is correct. By Le Chatelier's principle, an increase in temperature favors the endothermic direction (forward reaction) to absorb added heat. This increases the concentration of products and decreases the concentration of reactants at equilibrium. Because $K_c = \frac{[\text{products}]}{[\text{reactants}]}$, the numerical value of K_c increases.	[1]
103	D	Option D is correct. $K_c = \frac{[\text{NH}_3]^2}{([\text{N}_2][\text{H}_2]^3)}$. Units = $\frac{(\text{mol dm}^{-3})^2}{[(\text{mol dm}^{-3}) \times (\text{mol dm}^{-3})^3]} = \frac{(\text{mol dm}^{-3})^2}{(\text{mol dm}^{-3})^4} = (\text{mol dm}^{-3})^{-2} = \text{mol}^{-2} \text{dm}^6$.	[1]
104	B	Option B is correct. A conjugate base is formed when a Brønsted-Lowry acid donates one proton (H^+). Removing H^+ from H_2PO_4^- yields HPO_4^{2-} .	[1]
105	A	Option A is correct. Because the forward reaction is exothermic, low temperatures favor a high percentage equilibrium yield of NH_3 . However, at low temperatures, the rate of reaction is unacceptably slow because few reactant molecules possess the activation energy. 400–450 °C represents an optimized economic compromise that achieves a sufficient reaction rate in the presence of an iron catalyst while maintaining a viable equilibrium yield (~15%).	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
106	C	Option C is correct. Concentrations in 2.0 dm ³ : [SO ₂] = 0.40 / 2.0 = 0.20 mol dm ⁻³ ; [O ₂] = 0.20 / 2.0 = 0.10 mol dm ⁻³ ; [SO ₃] = 0.80 / 2.0 = 0.40 mol dm ⁻³ . $K_c = \frac{[\text{SO}_3]^2}{[\text{SO}_2]^2 \times [\text{O}_2]} = \frac{(0.40)^2}{(0.20)^2 \times 0.10} = 0.16 / 0.0040 = 40 \text{ dm}^3 \text{ mol}^{-1}$.	[1]
107	B	Option B is correct. $K_p = \frac{p(\text{NH}_3)^2}{p(\text{N}_2) \times p(\text{H}_2)^3} = \frac{(12.0)^2}{20.0 \times (60.0)^3} = 144 / [20.0 \times 216000] = 144 / 4320000 = 3.33 \times 10^{-5} \text{ kPa}^{-2}$.	[1]
108	D	Option D is correct. On the reactant side, there are 1 + 1 = 2 moles of gas. On the product side, there are 2 moles of gas. Because $\Delta n_{\text{gas}} = 0$, changing the pressure or volume changes the concentration of reactants and products by identical factors, causing no shift in equilibrium position.	[1]
109	A	Option A is correct. Both acids are monoprotic and contain identical moles of acid: $n = 0.0250 \text{ dm}^3 \times 0.10 \text{ mol dm}^{-3} = 0.0025 \text{ mol}$ of H ⁺ available for neutralisation. As OH ⁻ is added, it reacts with H ⁺ ; by Le Chatelier's principle, CH ₃ COOH continuously dissociates until all acid is neutralised. Therefore, both require exactly 25.0 cm ³ of 0.10 mol dm ⁻³ NaOH.	[1]
110	C	Option C is correct. In this reaction, an H ⁺ ion (proton) is transferred directly from an HCl molecule to the non-bonding lone pair on the nitrogen atom of an NH ₃ molecule to form the ammonium cation, NH ₄ ⁺ , and chloride anion, Cl ⁻ . By Brønsted-Lowry definition, HCl acts as a proton donor (acid) and NH ₃ acts as a proton acceptor (base).	[1]